

## 1. Customers Dataset

**File Name:** olist\_customers\_dataset.csv

### Purpose

Stores customer location information.

### What does each row represent?

One customer.

### Primary Key

customer\_id

### Foreign Keys

None

### Connected Tables

- Orders (customer\_id)

### Relationship

One customer → Many orders

### Columns

| Column                   | Description                                                                  |
|--------------------------|------------------------------------------------------------------------------|
| customer_id              | Unique identifier for each customer                                          |
| customer_unique_id       | Unique identifier representing the actual customer across multiple purchases |
| customer_zip_code_prefix | Customer ZIP code prefix                                                     |
| customer_city            | Customer city                                                                |
| customer_state           | Customer state                                                               |

### Business Questions

- Which state has the most customers?
  - Which city has the highest number of customers?
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## 2. Orders Dataset

**File Name:** olist\_orders\_dataset.csv

### Purpose

Stores order lifecycle information from purchase to delivery.

### What does each row represent?

One order placed by a customer.

### Primary Key

order\_id

### Foreign Key

customer\_id

### Connected Tables

- Customers
- Order Items
- Payments
- Reviews

### Relationship

- One customer → Many orders
- One order → Many order items
- One order → Many payments
- One order → One review (generally)

### Columns

| Column      | Description                   |
|-------------|-------------------------------|
| order_id    | Unique order ID               |
| customer_id | Customer who placed the order |

| Column                        | Description                 |
|-------------------------------|-----------------------------|
| order_status                  | Current status of the order |
| order_purchase_timestamp      | Purchase date and time      |
| order_approved_at             | Payment approval time       |
| order_delivered_carrier_date  | Date handed to carrier      |
| order_delivered_customer_date | Actual delivery date        |
| order_estimated_delivery_date | Expected delivery date      |

### Business Questions

- Monthly order trends
- Delivery performance
- Order status distribution
- Delivery delays

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### 3. Order Items Dataset

**File Name:** olist\_order\_items\_dataset.csv

#### Purpose

Stores details of every product included in an order.

#### What does each row represent?

One product within an order.

#### Primary Key

Composite Key:

- order\_id
- order\_item\_id

#### Foreign Keys

- order\_id

- product\_id
- seller\_id

### Connected Tables

- Orders
- Products
- Sellers

### Relationship

- One order → Many items
- One product → Many order items
- One seller → Many order items

### Columns

| Column              | Description                  |
|---------------------|------------------------------|
| order_id            | Order ID                     |
| order_item_id       | Item number within the order |
| product_id          | Purchased product            |
| seller_id           | Seller of the product        |
| shipping_limit_date | Shipping deadline            |
| price               | Product price                |
| freight_value       | Shipping cost                |

### Business Questions

- Top-selling products
  - Revenue by seller
  - Shipping cost analysis
-

#### 4. Products Dataset

**File Name:** olist\_products\_dataset.csv

**Purpose**

Stores product information.

**What does each row represent?**

One product.

**Primary Key**

product\_id

**Foreign Keys**

None

**Connected Tables**

- Order Items
- Category Translation

**Relationship**

One product → Many order items

**Columns**

| Column                     | Description            |
|----------------------------|------------------------|
| product_id                 | Product ID             |
| product_category_name      | Category in Portuguese |
| product_name_length        | Product name length    |
| product_description_length | Description length     |
| product_photos_qty         | Number of photos       |
| product_weight_g           | Weight (grams)         |
| product_length_cm          | Length                 |

| Column | Description |
|--------|-------------|
|--------|-------------|

|                   |        |
|-------------------|--------|
| product_height_cm | Height |
|-------------------|--------|

|                  |       |
|------------------|-------|
| product_width_cm | Width |
|------------------|-------|

#### Business Questions

- Best-selling categories
- Product size analysis
- Weight distribution

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### 5. Sellers Dataset

**File Name:** olist\_sellers\_dataset.csv

#### Purpose

Stores seller information.

#### What does each row represent?

One seller.

#### Primary Key

seller\_id

#### Foreign Keys

None

#### Connected Tables

- Order Items

#### Relationship

One seller → Many order items

#### Columns

| Column | Description |
|--------|-------------|
|--------|-------------|

| Column | Description |
|--------|-------------|
|--------|-------------|

|           |           |
|-----------|-----------|
| seller_id | Seller ID |
|-----------|-----------|

|                        |          |
|------------------------|----------|
| seller_zip_code_prefix | ZIP code |
|------------------------|----------|

|             |             |
|-------------|-------------|
| seller_city | Seller city |
|-------------|-------------|

|              |              |
|--------------|--------------|
| seller_state | Seller state |
|--------------|--------------|

### Business Questions

- Top-performing sellers
- Sales by seller location

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## 6. Order Payments Dataset

**File Name:** olist\_order\_payments\_dataset.csv

### Purpose

Stores payment information for orders.

### What does each row represent?

One payment transaction.

### Primary Key

No single primary key in the raw dataset.

### Foreign Key

order\_id

### Connected Tables

- Orders

### Relationship

One order → One or more payment records

### Columns

| Column               | Description             |
|----------------------|-------------------------|
| order_id             | Order ID                |
| payment_sequential   | Payment sequence number |
| payment_type         | Payment method          |
| payment_installments | Number of installments  |
| payment_value        | Amount paid             |

### Business Questions

- Most used payment method
- Installment analysis
- Revenue by payment type

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## 7. Order Reviews Dataset

**File Name:** olist\_order\_reviews\_dataset.csv

### Purpose

Stores customer reviews for orders.

### What does each row represent?

One review for an order.

### Primary Key

review\_id

### Foreign Key

order\_id

### Connected Tables

- Orders

### Relationship

One order → One review (in most cases)



## Columns

| Column                 | Description        |
|------------------------|--------------------|
| review_id              | Review ID          |
| order_id               | Order ID           |
| review_score           | Rating (1–5)       |
| review_comment_title   | Review title       |
| review_comment_message | Review description |

## Business Questions

- Average review score
  - Review score distribution
  - Impact of delivery on ratings
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## 8. Geolocation Dataset

**File Name:** olist\_geolocation\_dataset.csv

### Purpose

Stores Brazilian geographical location data.

### What does each row represent?

One ZIP code location.

### Primary Key

No primary key.

### Foreign Keys

None

### Connected Tables

- Customers
- Sellers

### Relationship

Linked using ZIP code prefix.

### Columns

| Column                      | Description |
|-----------------------------|-------------|
| geolocation_zip_code_prefix | ZIP code    |
| geolocation_lat             | Latitude    |
| geolocation_lng             | Longitude   |
| geolocation_city            | City        |
| geolocation_state           | State       |

### Business Questions

- Geographic distribution
- Customer location mapping
- Seller location mapping

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## 9. Product Category Translation Dataset

**File Name:** product\_category\_name\_translation.csv

### Purpose

Translates Portuguese product categories into English.

### What does each row represent?

One product category translation.

### Primary Key

product\_category\_name

### Foreign Keys

None

### Connected Tables

- Products

### Relationship

One category → Many products

### Columns

| Column                        | Description         |
|-------------------------------|---------------------|
| product_category_name         | Portuguese category |
| product_category_name_english | English category    |

### Business Questions

- Display English category names in reports
- Category-wise sales analysis

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### Relationship Summary

| Parent Table | Child Table          | Common Column         | Relationship        |
|--------------|----------------------|-----------------------|---------------------|
| Customers    | Orders               | customer_id           | One-to-Many         |
| Orders       | Order Items          | order_id              | One-to-Many         |
| Orders       | Payments             | order_id              | One-to-Many         |
| Orders       | Reviews              | order_id              | One-to-One (mostly) |
| Products     | Order Items          | product_id            | One-to-Many         |
| Sellers      | Order Items          | seller_id             | One-to-Many         |
| Products     | Category Translation | product_category_name | Many-to-One         |
| Customers    | Geolocation          | ZIP code prefix       | Many-to-One         |
| Sellers      | Geolocation          | ZIP code prefix       | Many-to-One         |